



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

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Priority Queue Using Arrays

PRIORITY QUEUE USING ARRAYS

Write a C program to implement a Priority Queue using Arrays for processing emergency service requests. Requests with higher priority must be processed before lower-priority requests irrespective of arrival order. Sample Input: Insert (101, Priority 3), Insert (102, Priority 1), Insert (103, Priority 2), Delete. Sample Output: Processed Request: 102. Remaining Queue: 103 101.

PROGRAM EDITOR

C

✓ Updated

Run

Save

```
1 #include <stdio.h>
2
3 #define MAX 10
4
5 typedef struct {
6     int id;
7     int priority;
8 } Request;
9
10 Request pq[MAX];
11 int size = 0;
12
13 void insert(int id, int priority) {
14     if (size < MAX) {
15         pq[size].id = id;
16         pq[size].priority = priority;
17         size++;
18     } else {
19         printf("Queue Overflow\n");
20     }
21 }
```

OUTPUT

```
Processed Request: 102
Remaining Queue: 101 103
```

INPUT

Insert(101, Priority 3)

Insert(102, Priority 1)

Insert(103, Priority 2)

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